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ORIGINAL ARTICLE

The effect of 16 weeks gymnastic training on social skills and neuropsychiatric functions of autistic children

Azadeh Zamani Jam[1,2] · Rasoul Hemayat Talab[3] · Mahmoud Sheikh[3] · Farnaz Torabi[4] · Forouzan Rafie[5]

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Abstract

The aim of the present study was to evaluate the effect of 16 weeks of gymnastic exercises on motor and neuropsychological skills of children with autism. The statistical population consisted of autistic children in a charity center in Karaj. Thirty autistic children 8–12 years of age, who were eligible to be included in the study, were randomly selected and assigned to two test and control groups. To evaluate motor development, the short form of Bruininks–Oseretsky test and to evaluate neuropsychological characteristics, Conners scales were used. The test group carried out gymnastic exercises for 16 weeks (three sessions a week, with each session lasting for 45 min); the control group subjects were engaged in their daily activities during the period. Two-way ANOVA was used to compare the two groups. There was a significant difference in the total motor skill scores between the two groups ($P < 0.05$); however, there were no significant differences between the two groups in subscales such as the speed of running and agility, strength and the speed of reaction/response between the two groups ($P > 0.05$). On the other hand, all the neuropsychological characteristics improved except for language performance. It can be concluded from the results of the present study that gymnastic exercises might have an effective role in improving motor and neuropsychological skills of children with autism.

Keywords Autism · Gymnastic · Motor development · Neuropsychological characteristics

Introduction

Autism as a general term which comprises a spectrum of developmental disorder, affecting different aspects of psychological, neurological, gross and fine motor skills, communication skills and attention. Asperger syndrome or autism disorder with high function is at the extreme end of the spectrum [1, 2].

The prevalence of autism spectrum disorders has increased all over the world. Studies in 1960 reported a prevalence rate of 4 in 10,000 for the autism spectrum disorders; however, the American National Institute of Mental Health estimated a prevalence rate of 1:88 in 2012 [3].

Several studies have reported problems with attention, executive functions and learning in autistic children [4]. However, possibly attention problem in autistic children is not the only problem in such children. The neuropsychological skills are directly related to each other due to similarities in biologic and cerebral bases. Neuropsychological skills consist of some basic cognitive performances such as attention, executive performances, memory, meta-cognition and the speed of processing. Based on previous studies, these children have problems in several neuropsychological aspects [5], especially in attention and imitation [6], memory and cognitive [7] and learning language [8].

Izawa et al. found in their study that autistic individuals were motivated by impairment of motor skills, and concluded that their mimic defects caused learning disabilities to be impaired [9]. Naito et al. considered the damage to mind theory in these children to be related to linguistic damage and even prognostic skills [10]. Whalen et al., in another study that looked at four children with autism

✉ Forouzan Rafie
fr.rafie87@ut.ac.ir

1 University of Tehran, Alborz campus, Tehran, Iran

2 Payame Noor University, Tehran, Iran

3 Faculty of Physical Education and Sport Sciences, University of Tehran, Tehran, Iran

4 Faculty of Physical Education and Sport Sciences, Payame Noor University of Tehran, Tehran, Iran

5 Neuroscience Research Centre, Kerman University of Medical Sciences, Kerman, Iran

spectrum disorder during the 10 weeks, found that attention in these children had a positive and significant correlation with information processing, imitation, language skills and social skills [11].

Studies have reported at least three anomalies as basic defects in psychological disorders of autistic subjects: a defect in the theory of mind, a defect in executive functions, a defect in central coherence; in this context, executive functions consist of working memory, inhibition and flexible control of attention. Speech problems are one of the most common problems in such children; they might have delays in speaking or might incorrectly use their speech abilities due to a lack of proper understanding of words. For examples, they might repeat a word or a sentence that they are told (parrot-fashion repetition) or only use a series of repetitive words for communication [12–14].

Motor disturbances are another prominent characteristic of autism and many studies have been undertaken in this respect. In a study by Pitetti et al., 50–70% of autistic children exhibited a clear delay in basic motor skills [15]. In a study by Green et al., a prevalence rate of 79% was reported for motor disturbances in these children [16]. In addition, Ozonoff et al. evaluated gross motor skills in children with autism and reported a delay in the maturation of such skills in these children [17]. In addition, Reid and Staples evaluated autistic children with TGMD-2 test and reported defects and delays in motor skills in autistic children compared to healthy subjects [18]. Coordination defects in autistic children consisted of disturbances in programming of movement, executive of two-step skills and control of moment [19]. Lang et al. evaluated autistic children with high function and reported low motor skills in these children [20].

Inefficacy of motor skills results in aggravation of psychological and behavioral disorders. Neuropsychological disorders such as defects in executive function and attention in autistic children might persist in older ages, resulting in serious problems in doing school homework and carrying out personal affairs. Therefore, it is necessary to prepare early interventional measures to improve them.

In recent years, since sports activities have been useful role in the development of motor and cognitive skills in healthy children, it seems to be effective in decreasing behavioral and motor disturbances.

A defect in motor control and behavior inhibition might be attributed to a high level of excitation and abnormal responses of these children to sensory stimuli that decrease the executive levels and the ability for executive functions [20].

It might be concluded from all the above-mentioned findings that what mainly leads to defects in the theory of mind in autistic subjects is a defect in the necessary psychological inputs, which is brought about by problems with neurological mechanisms. Finally, if this theory is correct, it is expected that therapeutic and non-therapeutic interventions that are intended to improve psychological functions will be more effective.

Several studies have shown the effect of sports activities on improving autism spectrum disorders, especially motor disturbances [21], behavioral problems [22], and improvements in communication skills [23]. However, some studies have shown that bodily activities and exercises have had no effects on improving motor skills in children [24].

Considering discrepancies in the results of previous studies and an increase in the number of patients with autism in the community and also given the economic costs and severe motor and psychological problems in children with autism, the present study was undertaken to evaluate the effect of engaging in gymnastic activities for 16 weeks on motor development and neuropsychological characteristics of children with autism.

Materials and methods

In the present descriptive/analytical study, 30 children with autism spectrum disorders, who were eligible based on inclusion criteria, were randomly selected from autistic children in a charity organization in Karaj, Iran.

Inclusion criteria

Inclusion criteria consisted of autism with high function, an age range of 8–12 years, parental consent from, IQ ≥ 70, and absence of any motor and physical inabilities.

To determine the severity and the level of function of autistic children, High-function Autism Spectrum Screening Questionnaire (ASSQ) was used. The validity and reliability of the Persian version of this questionnaire were confirmed in 2011 by Kasechi et al. [25]. Cronbach's α-coefficient of this questionnaire was estimated at $r = 0.64$, indicating its reliability at an acceptable level. In this study, children who gained scores > 22 were considered high-function autistic children.

Determination of motor skill development

Development of children's motor skills was determined with the use of the short form of BOTMP-2 scale, which measures the development of motor skills in children 4–14 years of age and consists of eight subtests with 14 items. Four subtests determine gross motor skills (the speed of running and agility, balance, bilateral coordination and strength), three subtests determine fine motor skills (the speed of reaction/response, visual-motor control and the speed and agility of the upper limb) and one subtest determines both motor skills and coordination of the upper limb [26]. The reliability coefficient of the retest of the short form of this questionnaire has been estimated at 86% [27].

Determination of neuropsychological characteristics

Conners neuropsychological questionnaire was constructed by Conners in 2007 to evaluate neuropsychological skills, including attention, executive functions, sensory and motor functions, language functions, and memory and learning functions in two forms for parents and teachers [28]. In the present study the parents' versions was used. The validity of this questionnaire was evaluated in Iran by Abedi et al. [29]. The validity of the questionnaire was estimated at 90% using factor analysis technique and its reliability was estimated at 85% using Cronbach's alpha.

Procedure

After explaining the aim of the study and gaining informed consent forms, 30 children who were eligible based an inclusion criteria were included in the pretest. The demographic data of the subjects and their consent forms were carefully examined before being included in the pretest. The parents were reassured about the confidential nature of the demographic data and they were informed that they were able to leave the study at any stage they wanted to.

Conners questionnaire was submitted to the parents to evaluate neuropsychological characteristics. The ambiguities in the questions were explained and the questionnaires were completed in the same session. Then, Bruininks–Oseretsky pretest was carried out and the results were recorded. After 16 weeks in the pretest, the children were re-evaluated with the same test and questionnaire.

Exercises

After the pretest, the subjects in the experimental group carried out preliminary and basic gymnastic exercises (such as balance activities and learning how to walking and balance on a beam, strength training on parallel bars, running and jumping on the springboard into the foam pit, climbing a rope, bridge, handstands, forward/backward rolls, etc.) in 48 sessions (three sessions each week) for 16 weeks. Each session consisted of 10 min of warm-up exercises, 25 min of basic gymnastic exercises, and 10 min of cooling down. During this period, the control group subjects were engaged in their routine daily activities.

Statistical analysis

SPSS 18 was used in the present study to analyze descriptive data with descriptive statistical parameters, consisting of means and percentages. After evaluation of normality of data, two-way ANOVA was used for the analysis of hypotheses at a significance level of $P < 0.05$.

Results

Table 1 presents the demographic data based on the personal data recorded in the test and control groups. Comparison of data showed no significant differences in demographic variables between the two groups ($P > 0.05$).

Since the pretest did not reveal any significant differences between the two groups ($P > 0.05$). Based on the results presented in Table 2, there were no significant differences in the subscales of the speed of running and agility [F (1, 27) = 0.662, $P = 0.423$] effect size = 0.024 and, strength [F (1, 28) = 0.010, $P = 0.920$], effect size = 0.000, and reaction time coordination [F (1, 27) = 5.717, $P = 0.241$] effect size = 0.175 between the two groups. Although the differences between the two groups in some subjects were significant statistically such as in the subscales of balance [F (1, 27) = 3063.107, $P = 0.001$], effect size = 0.999, bilateral coordination [F (1, 27) = 1751.400, $P = 0.001$], effect size = 0.998, visual control of movement and speed, the agility and coordination of the upper extremity [F (1, 27) = 277.136, $P = 0.001$], effect size = 0.999.

As shown in Table 3, the scores achieved by the subjects in the control group were lower in all the subscales of Conners neuropsychological characteristics compared to the test group, except for the language functions which were not significantly different between the two groups.

Discussion

Children with extensive developmental disturbances of autism have many problems in psychological characteristics and motor skills. Therefore, improving these skills is an important component of therapeutic interventions for such children. The main aim of the present study was to evaluate neuropsychological and motor characteristics of children with autism after 16 weeks of gymnastic exercises.

The results of Bruininks–Oseretsky test showed that the total scores of motor skills of autistic children were much lower than the standard norms. Since autistic children have cerebral deficiencies, these deficiencies exert negative effects on their motor functions, resulting in motor deficiencies. Reid and Staples [18], and Green et al. [16],

Table 1 Demographic data in the two study groups

Group	Gender	Age (year)	Weight (kg)	Height (cm)
Test	8 boys/7 girls	10.20	35.51	134.97
Control	9 boys/6 girls	10.12	34.22	135.41

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Table 2 The results of statistical analyses of subscales of motor development

Variable	Test group mean score (SD)	Control group mean score (SD)	*df* (1, 2)	*F*	*P*	Effect size
Bilateral coordination	6.40 (4.2)	2.19 (2.1)	(1, 27)	1751.400	0.001	0.998
Balance	8.11 (2.3)	5.02 (1.5)	(1, 27)	3063.107	0.001	0.999
Speed of running and agility	3.53 (2.2)	3.10 (0.6)	(1, 27)	0.662	0.423	0.024
Strength	4.70 4 (0.7)	4.63 (2.0)	(1, 27)	0.010	0.920	0.000
Speed of reaction/response	21.0 (5.3)	20.80 (3.9)	(1, 27)	5.717	0.241	0.175
Agility and coordination of the upper limb	6.58 (1.4)	4.11 (0.2)	(1, 27)	277.136	0.001	0.999

Table 3 The results of statistical analyses of neuropsychological subscales

Variable	Test group mean score (SD)	Control group mean score (SD)	*df* (1, 2)	*P*	*F*	Effect size
Attention	26.9 (6.3)	38.26 (3.4)	(1, 27)	0.001	10,682.5	0.997
Sensory-motor function	18.51 (2.1)	30.12 (3.5)	(1, 27)	0.001	17,891.4	0.998
Language functions	24.03 (3.4)	24.12 (4.5)	(1, 27)	0.535	0.395	0.14
Memory and learning function	29.54 (1.5)	41.24 (4.6)	(1, 27)	0.001	2542.6	0.999
Executive functions	19.0 (2.4)	31.40 (5.1)	(1, 27)	0.001	2020.9	1.000
Ability and speed of cognitive processing	11.3 (6.4)	23.32 (3.2)	(1, 27)	0.001	168.224	0.871

reported that motor skills in children with autism are much lower than those in their healthy and normal peers.

Based on the results of Bruininks–Oseretsky in the present study, the total scores of motor skills of the experimental group were significantly higher than those of the control group, consistent with the results of studies by Fergusson et al. [22], Pitetti et al. [15], Fragla Pinkham et al. [30], and Rafie et al. [21]. Based on the reports of the researchers above and the results of the present study, it might be concluded that exercises and bodily activities result in the development and promotion of motor functions of these children. In other words, gymnastic exercises resulted in proper development of movement, control and coordination skills of children with autism, improving them.

However, subtests for the speed of running and agility and the speed of reaction/response did not reveal any significant differences between the two groups, and the bodily exercises used in the present study were unable to affect these skills in the subjects in the experimental group. The results of the present study were consistent with those of studies by Rafie et al. [21], Hameury [31], and Lochbaum [32], and contrary to the results of a study by Yilmaz et al. [33]. To explain these results, it might be pointed out that autistic children require longer times for execution of speedy tasks due to defects in motor processing, and due to defects in active memory they cannot retain data in their memory for designing and predicting exercises. Therefore, they face difficulties in time-dependent behaviors and execution of two-step and simultaneous tasks that require a rapid reaction and motor programming.

In addition, the results showed no significant differences in the mean scores of strength between the test and control groups. Since autistic children have poor muscular tonicity and muscular contraction [33], gymnastic exercises were unable to increase their muscular strength and tonicity, consistent with the results of a study by Fraglapinkham [30], who showed that exercise was unable to change strength and flexibility; however, these results are different from those reported by Yilmaz et al. [33], and Rafie et al. [21]. Such a discrepancy might be attributed to the age of the subjects, the number of exercise sessions and the nature and type of the exercises used.

On the other hand, evaluation of the results of Conners test and neuropsychological characteristics showed that autistic children's scores in the exercise group significantly increased in all the variables evaluated except for language function. In this context, the results of a study by Bass et al. showed that 12 weeks of horse riding improved attention and social skills in autistic children [34]. In addition, a study by Hameury et al. showed that horse riding resulted in improvements in attention (22%) and imitation skills (60%) in 5–7-year-old autistic children [31].

These results are consistent with those of a study by Reid et al., who reported that sports activities did not affect speech in three young subjects with autism [35]; in addition, these results are contrary to the results of a study by Yilmaz, who reported a decrease in echolalia in a 9-year-old autistic child after carrying out different exercises in water [33]. This might be explained by the fact that carrying out coordinated and group exercises of gymnastics resulted in an increase in the level of executive functions, attention, cognitive processing and sensory-motor function. Gymnastic exercises resulted in an increase in balance, flexibility and body control through various movements such as running, jumping and balance movements on one hand and on the other hand, children's imitation during group performance of gymnastic activities might serve as a factor affecting development of neuropsychological skills; however, due to the complexity of the process of production, understanding and use of words, gymnastic exercises were unable to improve language disorders of autistic children participating in the present study.

There are some limitations in this study, the low sample size that limits generalization therefore future research is suggested to establish the applicability of the current findings to more participants and higher sample size, and at different severity of the autistic disorders. Also, the differences in cognitive abilities and motor skills were not evaluated before experiment and might have influenced our findings. So is needed to attend the impact of these variables in further researches.

Conclusion

Considering the high prevalence of autism and behavioral, psychological and behavioral problems in such individuals and in their families as a result, early diagnosis of these disorders and their treatment appear to be necessary to prevent subsequent functional disorders. Considering the fact that autism is not a solitary disturbance and consists of a spectrum of motor, behavioral and social disorders, with many psychological effect on the life of the child and parents, it is necessary for specialists and authorities in this field to pay adequate attention to the positive effects of motor and sports exercises and refer these children to specialty centers that provide proper sports activities for these children. The results of the present study showed that gymnastics can engage various organs of the body and different aspects of motor activities, resulting in the necessary coordination between neurons and muscles, with a subsequent increase in balance, motor coordination and executive functions in autistic children.

Compliance with ethical standards

Conflict of interest None of the authors have financial or other conflicts of interest in regards to this research.

Ethical approval All procedures performed in this study involving human participants were in accordance with ethical standard of the Tehran University and, all the experimental protocols were approved by the Ethic Committee of Tehran University.

Informed consent Informed consent was obtained from all parents that their children participated in this study, and they have free to continue the protocol.

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